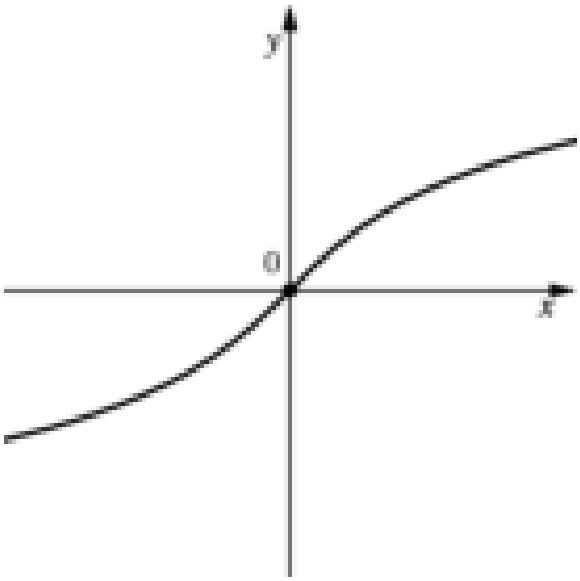
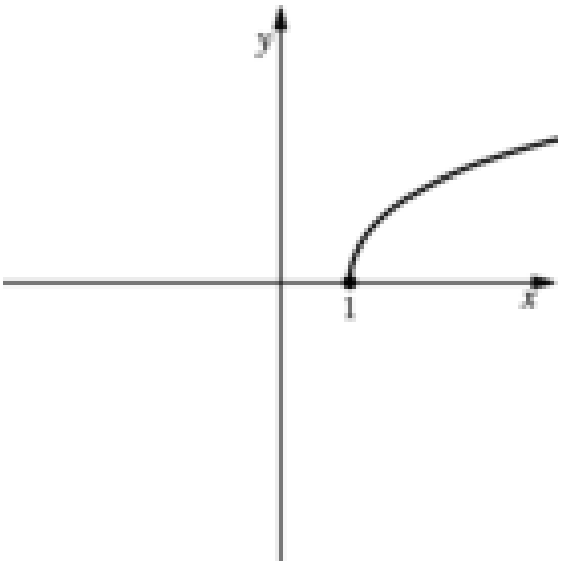
1.33 Graph Sketching: Inverse Hyperbolic Functions

Worked Solutions

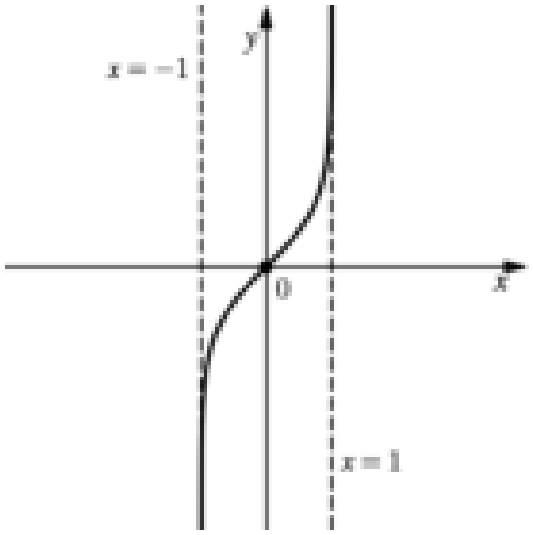
1 a)



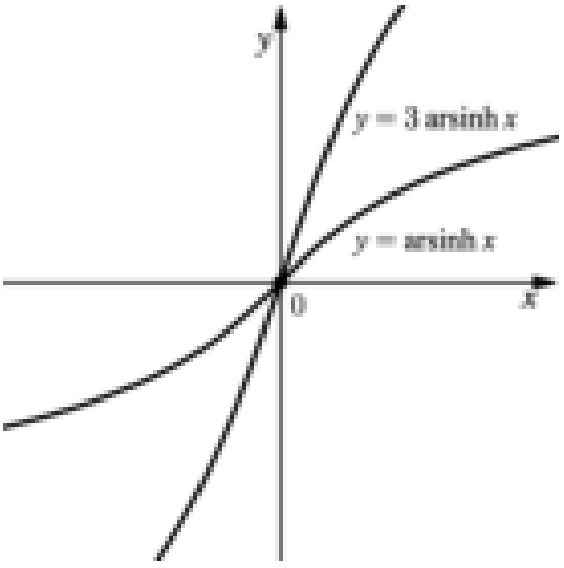
1 b)



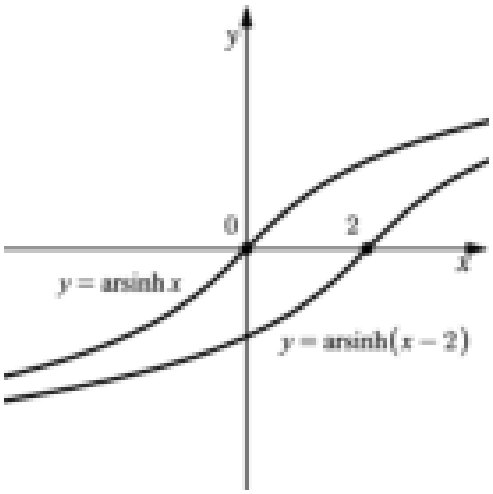
1 c)



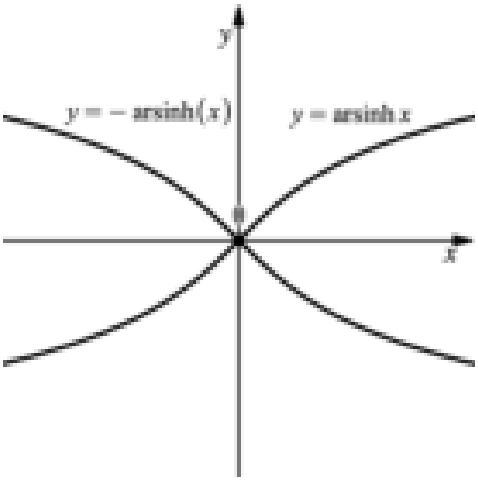
2 a)



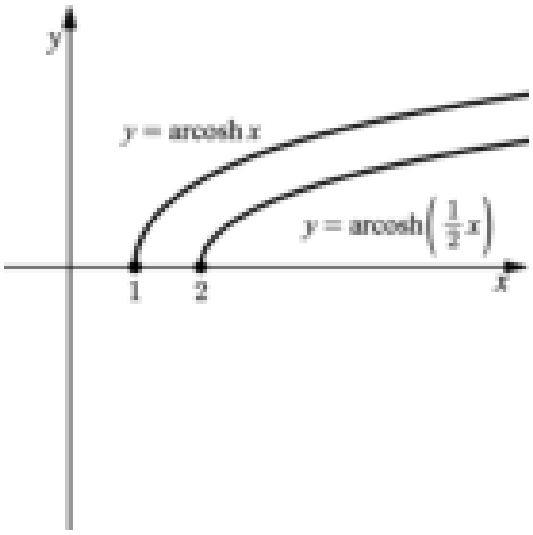
2 b)



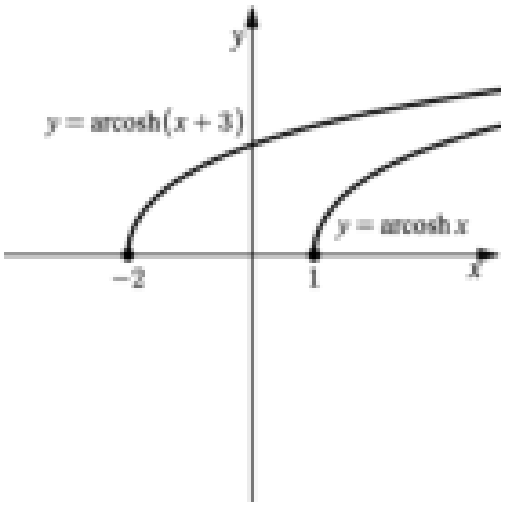
2 c)



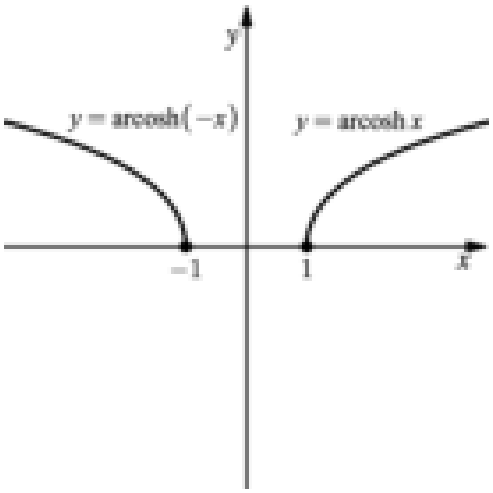
2 d)



2 e)



2 f)



3 a)

$y = \operatorname{arsinh} x + b$

$(-\frac{21}{20}, 0)$:

$$0 = \operatorname{arsinh}(-\frac{21}{20}) + b$$

$$\Rightarrow 0 = \ln(-\frac{21}{20} + \sqrt{(-\frac{21}{20})^2 + 1}) + b$$

$$\Rightarrow 0 = \ln(-\frac{21}{20} + \frac{29}{20}) + b$$

$$\Rightarrow b = -\ln\frac{2}{5} = \ln\frac{5}{2}$$

$$\therefore y = \operatorname{arsinh} x + \ln\frac{5}{2}$$

3 b)

$y = \operatorname{arsinh}(x - a)$

$(0, \ln(\frac{\sqrt{13}-3}{2}))$:

$$\ln(\frac{\sqrt{13}-3}{2}) = \operatorname{arsinh}(-a)$$

$$\Rightarrow \ln(\frac{\sqrt{13}-3}{2}) = \ln(-a + \sqrt{(-a)^2 + 1})$$

$$\Rightarrow \ln(\frac{\sqrt{13}-3}{2}) = \ln(-a + \sqrt{a^2 + 1})$$

$$\Rightarrow \frac{\sqrt{13}-3}{2} = -a + \sqrt{a^2 + 1}$$

$$\Rightarrow a + \frac{\sqrt{13}-3}{2} = \sqrt{a^2 + 1}$$

$$\Rightarrow (a + \frac{\sqrt{13} - 3}{2})^2 = a^2 + 1$$

$$\Rightarrow a^2 + (\sqrt{13} - 3)a + \frac{11 - 3\sqrt{13}}{2} = a^2 + 1$$

$$\Rightarrow (\sqrt{13} - 3)a + \frac{11 - 3\sqrt{13}}{2} = 1$$

$$\Rightarrow (\sqrt{13} - 3)a = \frac{3\sqrt{13} - 9}{2}$$

$$\Rightarrow a = \frac{3\sqrt{13} - 9}{2\sqrt{13} - 6} = \frac{3}{2}$$

$$\therefore y = \operatorname{arsinh}(x - \frac{3}{2})$$

4 a)

$y = \operatorname{arcosh} x + b$

$(\frac{13}{5}, 0)$:

$$0 = \operatorname{arcosh} \frac{13}{5} + b$$

$$\Rightarrow 0 = \ln(\frac{13}{5} + \sqrt{(\frac{13}{5})^2 - 1}) + b$$

$$\Rightarrow 0 = \ln(\frac{13}{5} + \frac{12}{5}) + b$$

$$\Rightarrow 0 = \ln 5 + b$$

$$\Rightarrow b = -\ln 5$$

$$\therefore y = \operatorname{arcosh} x - \ln 5$$

4 b)

$y = \operatorname{arcosh}(x - a)$

$(0, \ln(3 + 2\sqrt{2}))$:

$$\ln(3 + 2\sqrt{2}) = \operatorname{arcosh}(-a)$$

$$\Rightarrow \ln(3 + 2\sqrt{2}) = \ln(-a + \sqrt{(-a)^2 - 1})$$

$$\Rightarrow \ln(3 + 2\sqrt{2}) = \ln(-a + \sqrt{a^2 - 1})$$

$$\Rightarrow 3 + 2\sqrt{2} = -a + \sqrt{a^2 - 1}$$

$$\Rightarrow a + 3 + 2\sqrt{2} = \sqrt{a^2 - 1}$$

$$\Rightarrow (a + 3 + 2\sqrt{2})^2 = a^2 - 1$$

$$\Rightarrow a^2 + (6 + 4\sqrt{2})a + 12\sqrt{2} + 17 = a^2 - 1$$

$$\Rightarrow (6 + 4\sqrt{2})a + 12\sqrt{2} + 17 = -1$$

$$\Rightarrow (6 + 4\sqrt{2})a = -18 - 12\sqrt{2}$$

$$\Rightarrow a = -\frac{18 + 12\sqrt{2}}{6 + 4\sqrt{2}} = -3$$

$$\therefore y = \operatorname{arcosh}(x + 3)$$

5 a)

$y = \operatorname{artanh} x$ has horizontal asymptotes at $x = -1$ and $x = 1$.

$y = \operatorname{artanh} x \mapsto y = \operatorname{artanh}(x+5)-3$ is a translation by the vector $\begin{pmatrix} -5 \\ -3 \end{pmatrix}$, so the vertical asymptotes are translated 5 units to the left.

$\therefore y = \tanh(x+5) - 3$ has vertical asymptotes at $x = -6$ and $x = -4$.

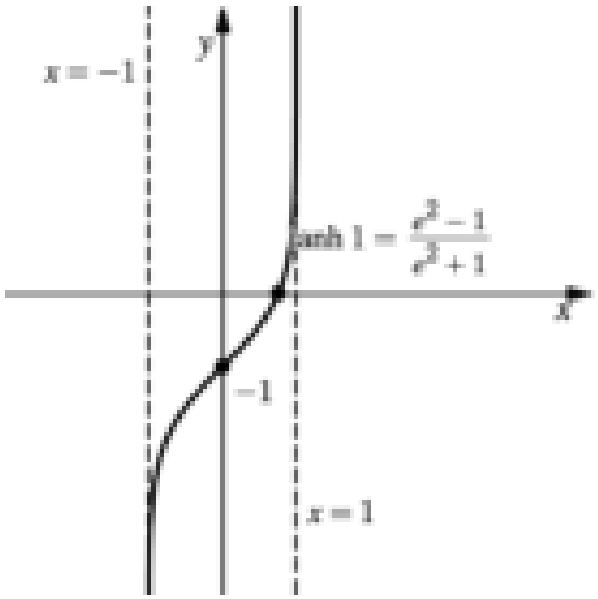
5 b)

$y = \operatorname{artanh} x$ has horizontal asymptotes at $x = -1$ and $x = 1$.

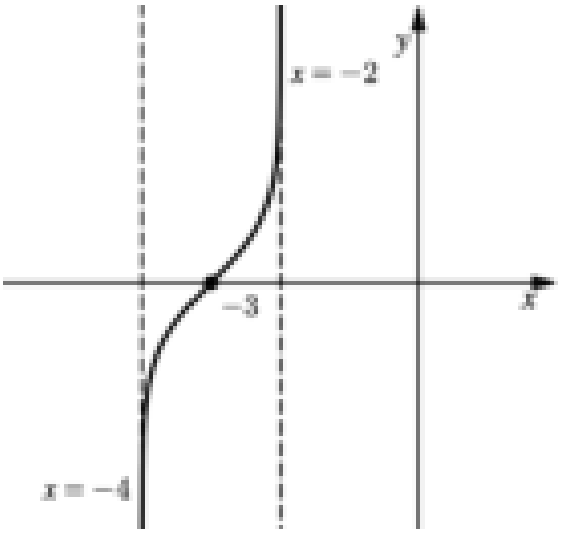
$y = \operatorname{artanh} x \mapsto y = \operatorname{artanh}(x-\frac{1}{3})+2$ is a translation by the vector $\begin{pmatrix} \frac{1}{3} \\ 2 \end{pmatrix}$, so the vertical asymptotes are translated $\frac{1}{3}$ units to the right.

$\therefore y = \tanh(x - \frac{1}{3}) + 2$ has vertical asymptotes at $x = -\frac{2}{3}$ and $x = \frac{4}{3}$.

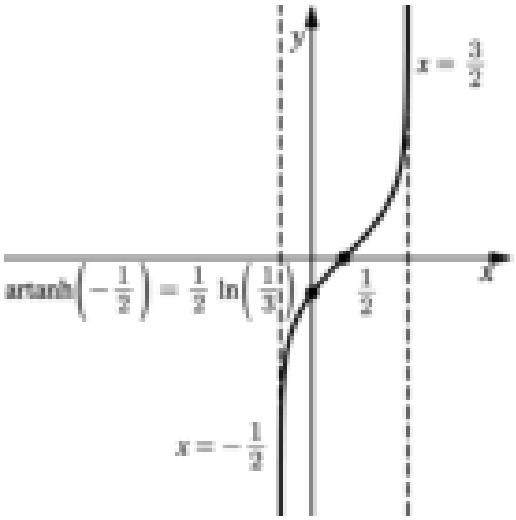
6 a)



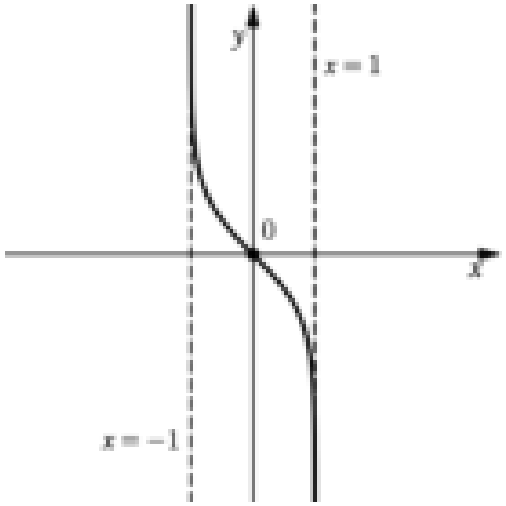
6 b)



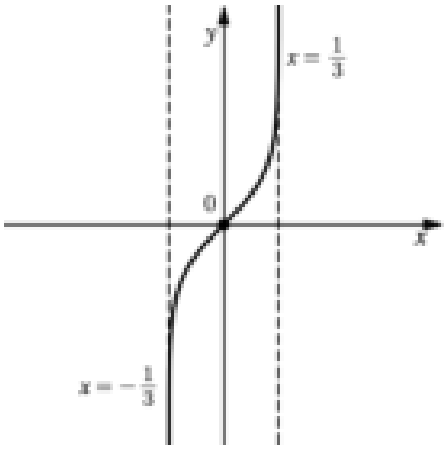
6 c)



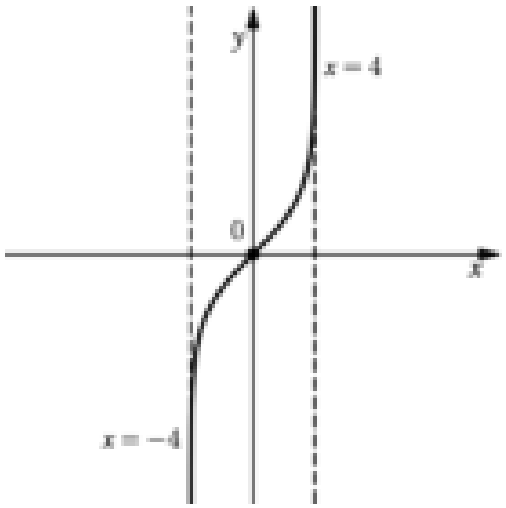
6 d)



7 a)



7 b)



8 a)

$\frac{-\frac{13}{8}+\frac{3}{8}}{2} = -\frac{5}{8}$

$y = \operatorname{artanh}(x + \frac{5}{8})$

8 b)

$y = \operatorname{artanh} x + b$

$(-\frac{4}{5}, 0)$:

$$0 = \operatorname{artanh}(-\frac{4}{5}) + b$$

$$\Rightarrow 0 = \frac{1}{2}\ln\frac{1+(-\frac{4}{5})}{1-(-\frac{4}{5})} + b$$

$$\Rightarrow 0 = \frac{1}{2}\ln\frac{1+(-\frac{4}{5})}{1-(-\frac{4}{5})} + b$$

$$\Rightarrow 0 = \frac{1}{2}\ln\frac{1}{9} + b$$

$$\Rightarrow 0 = \ln\frac{1}{3} + b$$

$$\Rightarrow b = -\ln\frac{1}{3} = \ln 3$$

$$\therefore y = \operatorname{artanh} x + \ln 3$$

8 c)

$y = \operatorname{artanh}(x - a)$

$(0, \ln(\frac{\sqrt{2}}{2}))$:

$$\ln(\frac{\sqrt{2}}{2}) = \operatorname{artanh}(-a)$$

$$\Rightarrow \ln(\frac{\sqrt{2}}{2}) = \frac{1}{2}\ln\frac{1+(-a)}{1-(-a)}$$

$$\Rightarrow 2\ln(\frac{\sqrt{2}}{2}) = \ln\frac{1-a}{1+a}$$

$$\Rightarrow \ln\frac{1}{2} = \ln\frac{1-a}{1+a}$$

$$\Rightarrow \frac{1}{2} = \frac{1-a}{1+a}$$

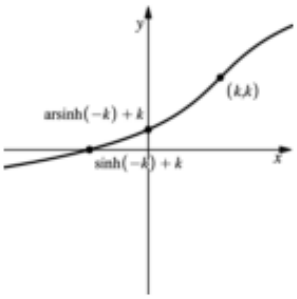
$$\Rightarrow 1+a = 2-2a$$

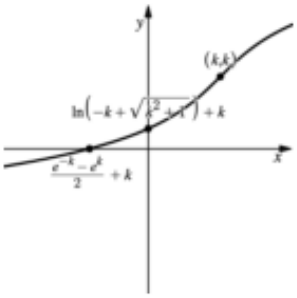
$$\Rightarrow 3a = 1$$

$$\Rightarrow a = \frac{1}{3}$$

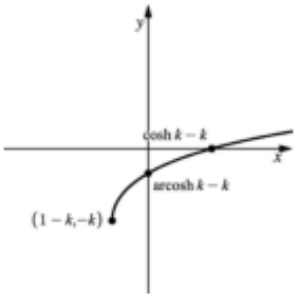
$$\therefore y = \operatorname{artanh}(x - \frac{1}{3})$$

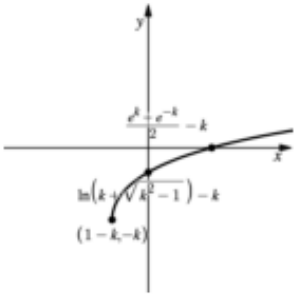
9 a)



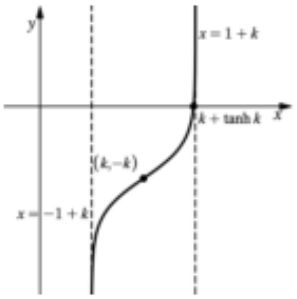


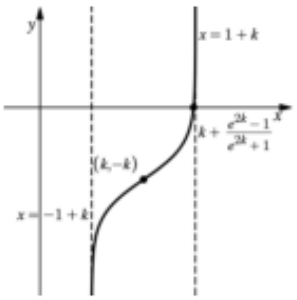
9 b)


y
cosh k − k
x
arcosh k − k
(1 − k, −k)

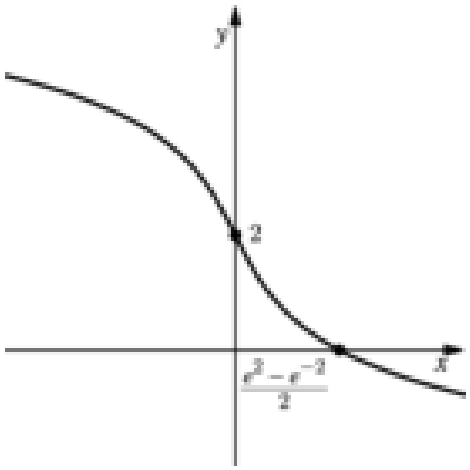

y
$\frac{e^k - e^{-k}}{2} - k$
x
$\ln\left(k + \sqrt{k^2 - 1}\right) - k$
(1 − k, −k)

9 c)

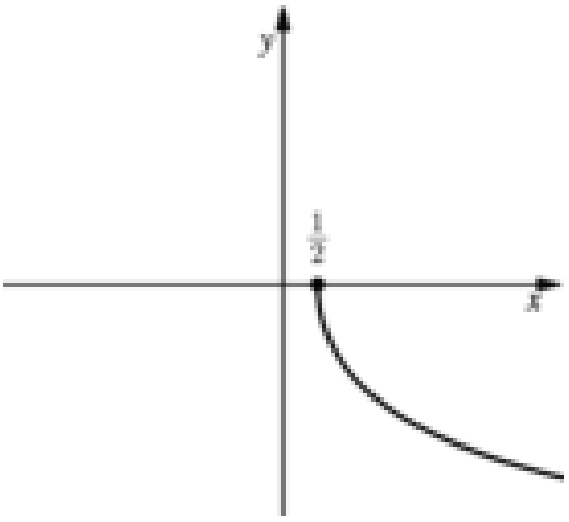

y
x = 1 + k
k + tanh k
x
(k, −k)
x = −1 + k


y
x = 1 + k
$k + \frac{e^{2k} - 1}{e^{2k} + 1}$
x
(k, −k)
x = −1 + k

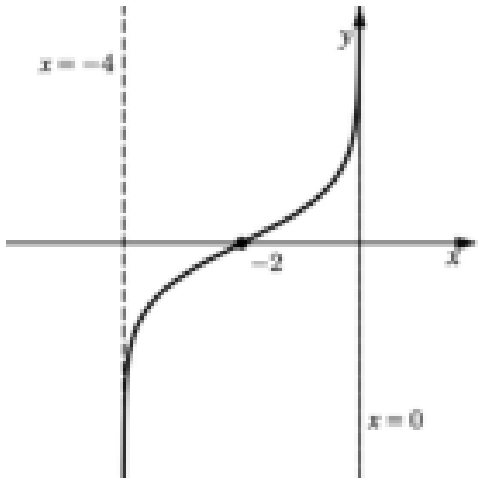
10 a)



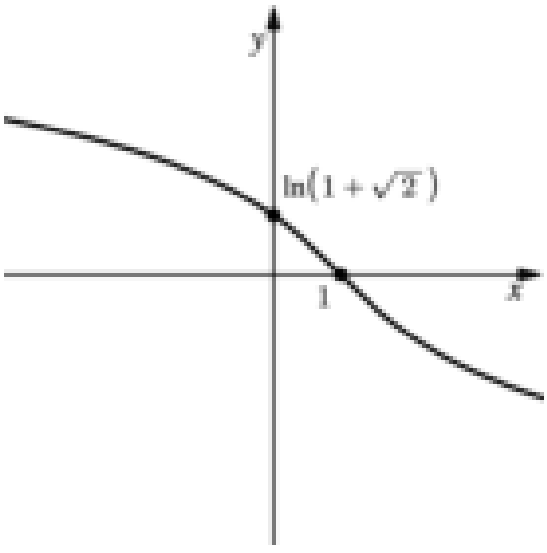
10 b)



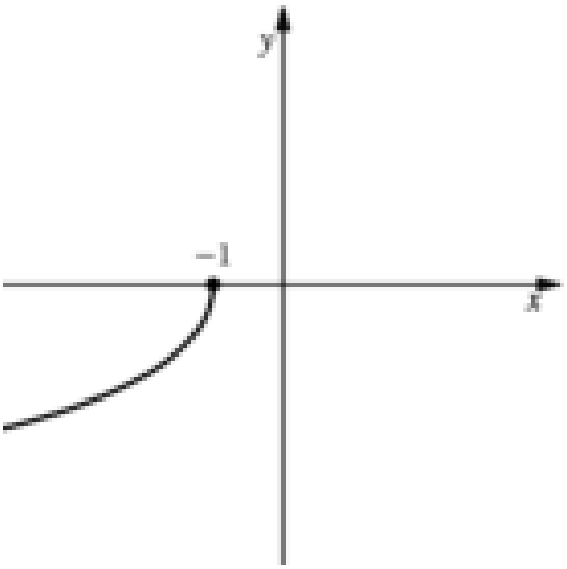
10 c)



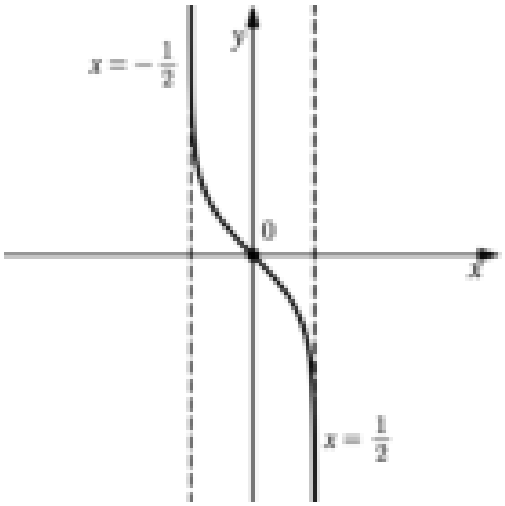
10 d)



10 e)



10 f)



11 a)

First consider $y = \operatorname{arcosh} x$:

Translate by $\begin{pmatrix} -k \\ -2k \end{pmatrix}$ to get $y = \operatorname{arcosh}(x + k) - 2k$:

But this isn't quite right as the x-intercept should be $\frac{e^2+e^{-2}}{2} - k$ instead of $\frac{e^{2k}+e^{-2k}}{2} - k$.

So start again with $y = \operatorname{arcosh} x$:

Then stretch, parallel to the y-axis, by scale factor k to get $y = k \operatorname{arcosh} x$:

Translate by $\begin{pmatrix} -k \\ -2k \end{pmatrix}$ to get $y = k\,\text{arcosh}(x+k) - 2k$:

$\therefore y = k\,\text{arcosh}(x+k) - 2k$.

11 b)

First consider $y = \text{arcosh}\,x$:

Reflect in the x-axis to get $y = -\,\text{arcosh}\,x$:

Translate by $\begin{pmatrix} k \\ k \end{pmatrix}$ to get $y = -\,\text{arcosh}(x-k) + k$:

But this isn't quite right as the x-intercept should be $\frac{e+e^{-1}}{2} - k$ instead of $\frac{e^k+e^{-k}}{2} - k$.

So start again with $y = \text{arcosh}\,x$:

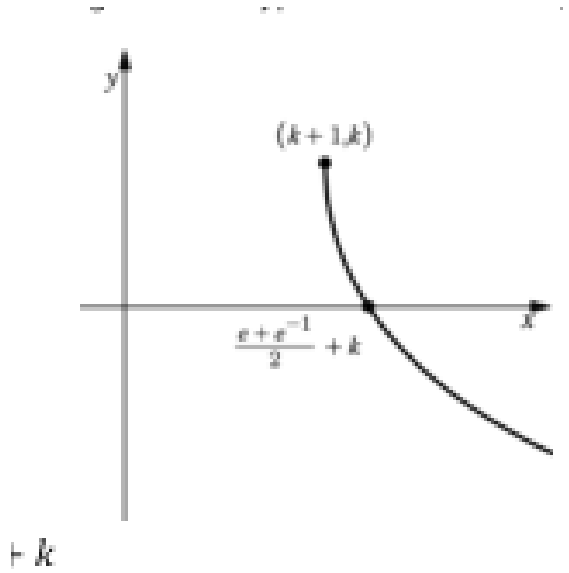
Then stretch, parallel to the y-axis, by scale factor k to get $y = k\,\text{arcosh}\,x$:

Reflect in the x-axis to get $y = -k\,\text{arcosh}\,x$:

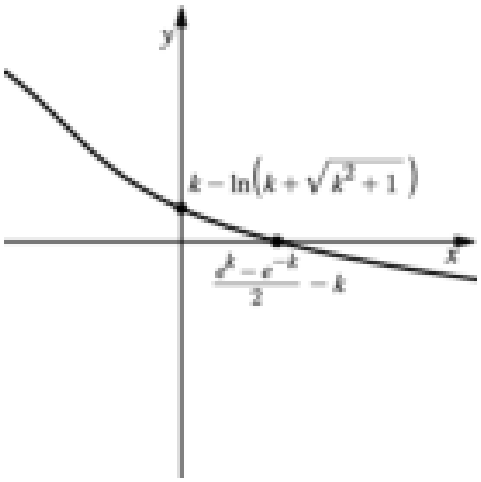
Translate by $\begin{pmatrix} k \\ k \end{pmatrix}$ to get $y = -k\,\text{arcosh}(x-k) + k$:

$\therefore y = -k\,\text{arcosh}(x-k) + k$.

12 a)

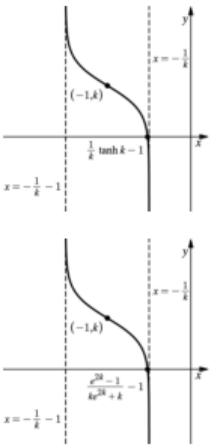


12 b)


y
$k - \ln\left(k + \sqrt{k^2+1}\right)$
x
$\frac{e^k - e^{-k}}{2} - k$


y

12 c)


y
$x=-\frac{1}{k}$
$(-1,k)$
$\frac{1}{k}\tanh k - 1$
x
$x=-\frac{1}{k}-1$
y
$x=-\frac{1}{k}$
$(-1,k)$
$\frac{e^{2k}-1}{ke^{2k}+k}-1$
x
$x=-\frac{1}{k}-1$